Program Evaluation and Review Technique (PERT) chart

Mitigation and Remediation planning tool

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Definition

- It defines the timing of a project, the dependencies between activities and enables the identification of the critical path.
- This is a project management tool used to plan, schedule, and coordinate tasks within a project by visually mapping activities and their dependencies.
- It describes a method for planning and controlling projects in the form of a network plan. Within this method, the PERT diagram is used as a graphical tool.
- Special symbols and arrows are used for creation to represent activities, milestones, and dependencies between tasks. Basic elements of the PERT diagram will include the following;
- PERT Charts are especially valuable for complex projects where timelines are difficult to estimate and precise scheduling is crucial for success. Mitigation and Remediation plans for IT projects very much use PERT Charts

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Benefits of a PERT Chart

- The PERT diagram offers several advantages within project management the following are some cited examples;

1. **Visualization of dependencies:** PERT Chart diagrams enable a clear visualization of the dependencies between tasks and activities in the project. This facilitates understanding of the order in which activities must be carried out.
2. **Critical path identification:** An important advantage of PERT networks is that they show the critical path in the project. This is the sequence of activities that determines the longest duration in the project. By focusing on the critical path, a project manager can identify bottlenecks and concentrate on minimizing delays
3. **Effective time management:** PERT diagram support time management by presenting the project schedule in clearly defined activities and their temporal dependencies. This helps a project manager to plan and allocate resources exactly where they are needed.

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Benefits of a PERT Chart

4. **Risk management:** By identifying dependencies and estimating activity times, the diagram uncovers potential risks. In this way, you can better respond to uncertainties and unforeseen events.

5. **Clear communication:** PERT diagrams provide a clear and easily unherdable representation of the project communication and thus improves team collaboration and cooperation with stakeholders.

6. **Resource optimization:** By elucidating the temporal requirements and dependencies between activities, the diagram ensure efficient management of resources.

7. **Project monitoring and control:** PERT diagrams serve as a reference point for monitoring project progress. If you need to make updates, their impact on status and schedule will become immediately apparent

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Challenges of PERT Charts

- Every project management tool has its own limitations. The following are some pitfalls of PERT diagrams;

1. The accuracy of the PERT chart depends heavily on the quality of the estimates for the activity times. As a project manager, you therefore, need to provide as realistic time estimates as possible. Incorrect estimate times leads to distorted schedule that does not meet the actual requirements of the project. The difficulty lies in adequately assessing uncertainties and unforeseeable events
2. The complexity of PERT diagrams is often underestimated in practice. This applies especially to extensive projects and people who are not familiar with the method. The multitude of activities , dependencies and events can make the diagram potentially confusing, which in turn can lead to misunderstanding and difficulties in interpretation. Therefore, it is important to train team members and stakeholders with regard to the meaning of the various elements in the PERT model. This ensures that all participants can effectively utilize the tool.

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Key features

- **Activities (Nodes):** the various tasks in the project are represented by nodes. These are visual blocks representing project tasks
- **Arrows:** Indicates the dependencies between activities. They indicate which activity must take place before or after another.
- **Time Estimates per task:** Each activity is provided with estimated times for its completion.
- **Optimistic time (O):** Shortest duration if everything goas perfectly.
- **Most likely time (M):** Expected duration under normal conditions.
- **Pessimistic time (P):** Longest possible duration if challenges occur.
- **Events:** milestones or points where several activities come together or diverge
- **Critical path:** The longest path through the network, which determines the shortest time for project completion.

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Key features

- **Critical path:** The longest sequence of dependance tasks. Delays along the critical path directly extend the entire project duration.
- **Milestones:** Key points when significant progress or deliverables are reached.
- Software for developing PERT Charts include the following;

1. **LucidChart** (free version available)
2. **Venngage** (free version available)
3. **SmartDraw** (No free version)
4. **EdrawMax** (Trial version)
5. **Creately** (Limited version)
6. **Microsoft Visio** (Advanced software – No free version but can work with Microsoft 365)
7. **Miro** and **Visme** are also web based software to create Mitigation and remediation projects.

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Creating a PERT Chart

- **List all Tasks:** Write down every activity needed for your project
- **Determine Task Dependencies:** Identify which activities must precede others.
- **Estimate time for each task:** Assign optimistic, most likely, and pessimistic durations.
- **Build the network diagram:** Draw nodes for activities, connected by arrows that illustrate dependencies.
- **Calculate the Critical path:** Follow the longest duration path through the network to determine the minimum project completion time.
- Remember that nodes are simply circles or rectangles which display a project activity.
- Arrows connect nodes, indicating the order and dependencies of tasks.
- The network may branch with parallel activities or merge when tasks depend on multiple prior completions.

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Examples for working with a PERT Chart

1. Construction of a network diagram (PERT diagram) from the list of activities and dependencies below. [5 Marks]

FLOATS (SLACK) in a Project

- In a PERT chart, several types of floats also known as slack are used to determine scheduling flexibility for each activity. Below are the typical types of floats (Slack) and what they mean in the project;

1. **Total Float (TF):** is the amount of time an activity can be delayed without delaying the project's completion time or date.

Formula: Total Float (TF) = Latest Finish (LF) – Earliest Finish (EF) or equivalently, **Total Float (TF) = Latest Start (LS) – Earliest Start (ES)**

- You need to first **calculate ES, EF (Forward Pass), and LS, LF (backward pass)** for each activity to get these values

FLOATS (SLACK) in a Project

2. **Free Float (FF):** is the amount an activity can be delayed without delaying the start of any immediate successor activity

Formula: Free Float (FF) = Earliest Start (ES) of successor activity – Earliest Finish (EF) of current activity or simply;

Free Float (FF) = ES (Successor) – EF (current)

Note that if multiple successors, use the earliest among them.

FLOATS (SLACK) in a Project

3. **Independent Float (IND F):** is the amount an activity can be delayed, assuming all predecessor activities finish as late as possible, and all successor activities start as early as possible. It can be thought as how much the activity can be delayed when both its predecessors and successors are constrained as strictly as possible.

Formula: Independent Float = Free Float – Slack of the Tail event typically, **Independent Float = ES (Successor) – LF (Tail event) – Duration of activity**

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FLOATS (SLACK) in a Project

4. **Interference Float (INT F):** is the part of total float which, if used, will delay the early start of any subsequent activity or consume float on the path. It shows all the portion of total float which, if used up, interferes with the float of the successors

Formula: Int Float = Total Float – Free Float

FLOATS (SLACK) in a Project summary

Float type	Meaning	Formula
Total Float	Delay possible without changing project completion	LF – EF or LS – ES
Free Float	Delay possible without delaying direct successor	ES (successor) – EF (Current)
Independent Float	Delay possible without affecting predecessors' late finish or successors' start	Free Float Slack of tail event
Interference Float	Portion of total float interfering with successors	Total Float – Free Float

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Examples of on calculating the floats of the project

- **Make reference to the discussions in class**

Estimates used in PERT

- The following are types of estimates used in PERT

1. **Optimistic also known as Least time estimate (O)** happens to be the shortest time required assuming everything goes well, no risks or issues. For example, all resources in terms of money and skilled labour and software required are readily available. All workers involved in the project also report on time.

Estimates used in PERT

2. **Pessimistic also known as Greatest time estimate (tp) or (P)** is the longest time assuming all possible problems occur. For example, skilled labour is available but very demotivated. Resources such as software needed, and money to complete the project are scarce.

3. **Most likely time estimate (tm) or (M)** is the realistic duration accounting for normal risks and issues.

Estimates used in PERT – useful formulas

- **Variance** of the activity time: $\boldsymbol{\sigma^2 = \left(\frac{P - O}{6}\right)^2}$

Standard deviation $\boldsymbol{\sigma = \frac{P - O}{6}}$

- Calculating the weighted average in PERT

Expected time = $\boldsymbol{O + 4M + P/6}$

- Where O is the optimistic estimate (best case scenario)
- M is the most likely estimate (the most realistic case)
- P is the pessimistic estimate (worst case scenario)

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Estimates used in PERT – useful formulas

- The formula above weights the most likely estimate four (4) times more than the optimistic and pessimistic estimates. The division by 6 normalizes the sum, because there are effectively **6 weights total (1 + 4 + 1)**
- The reasoning behind the weighted average is to reflect the beta distribution shape, where the most likely estimate is more probable, creating a balanced statistically grounded expected duration rather than a simple arithmetic average

Estimates used in PERT – Examples

- Consider the following examples for illustrations of the issues discussed above